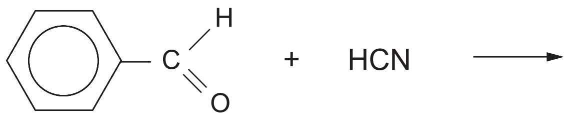
Examiner only

SECTION A

*Answer **all** questions in the spaces provided.*

1. *(a)* Complete the equation below, which shows the reaction of benzaldehyde with hydrogen cyanide. [1]



(b) Give the **formula** of the nucleophile that takes part in the reaction shown in part *(a)*. [1]

....

2. Decarboxylation occurs when a carboxylic acid is heated with sodalime.

State the name of the carboxylic acid that will produce propane when strongly heated with sodalime. [1]

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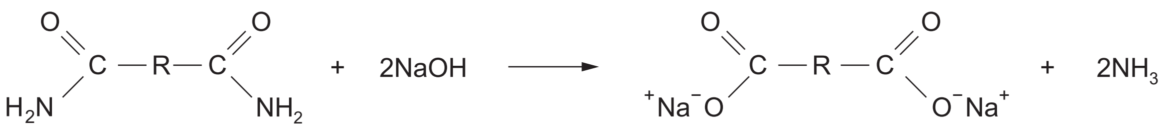
3. Ammonia is produced when an amide is heated with aqueous sodium hydroxide.



amide **T**

In an experiment 3.90 g of the aliphatic amide **T** gave 0.060 mol of ammonia.
Use this information to find the formula of the hydrocarbon group R, present in the amide **T**. [2]

Formula ....

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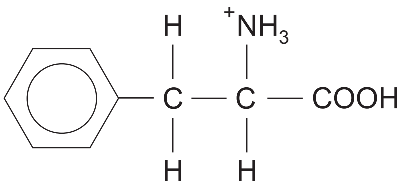
4. *(a)* In acid solution the formula of the ion formed from phenylalanine is



Give the formula of the ion formed from phenylalanine in basic solution. [1]

(b) The melting temperatures of four compounds are shown in the table.

Compound	Melting temperature / °C
CH_3CH_2COOH	–21
$CH_3CH(OH)COOH$	18
$CH_3CHBrCOOH$	26
$CH_3CH(NH_2)COOH$	289

Explain why the melting temperature of 2-aminopropanoic acid is very much higher than the other acids in the table. [2]

....

....

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5. On detonation, the explosive RDX produces carbon monoxide, steam and nitrogen.

$$C_3H_6N_6O_6(s) \longrightarrow 3CO(g) + 3H_2O(g) + 3N_2(g)$$

RDX
M_r 222

Calculate the mass of RDX that would give 1.00 m^3 of gaseous products measured at 100 °C and at 1 atm. Give your answer to an **appropriate** number of significant figures. [2]

(1 mol of any gas occupies a volume of 30.6 dm^3 under these conditions)

Mass of RDX = ... g

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